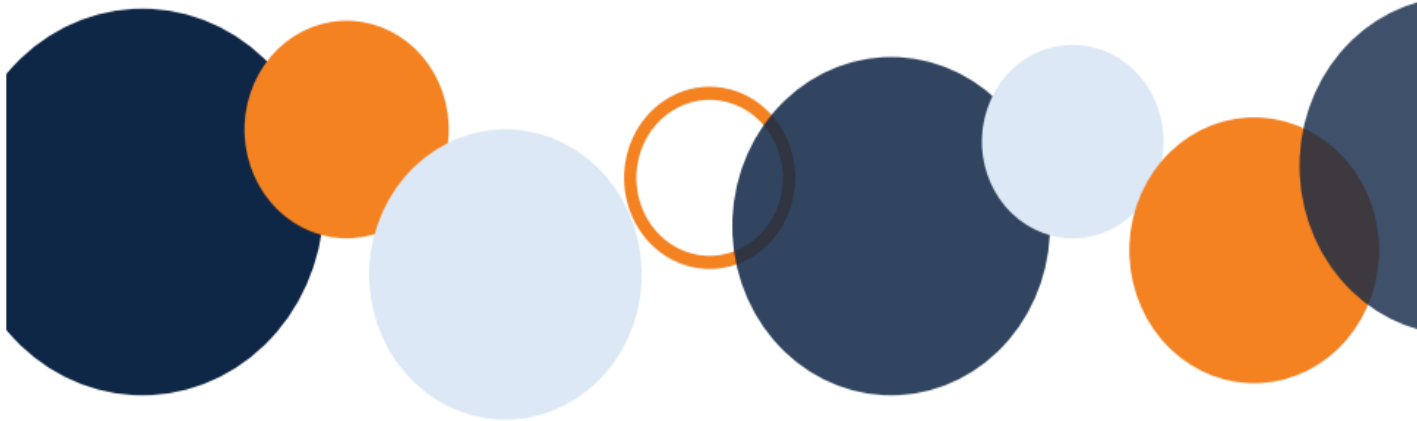


Who Won't Come Back, and Where Retention Spend Actually Pays Off

A churn-prediction and value-based targeting analysis of 232,079 shoppers across five months of a cosmetics e-commerce store.

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Executive Summary - Retention is the binding constraint, and the exposure is concentrated

Business problem: This cosmetics e-commerce store sells low-ticket items (median item price \$4.11), so value depends on customers returning, not on any single order. Yet 77.1% of the 232,079 shoppers active in October 2019 recorded no activity at all over the following two months - the observed non-return rate. By the time inactivity is visible, the customer is already gone, so the store needs to identify likely non-returners from early behavior and decide where limited retention budget should go.

Core findings: A churn-risk model built only on October behavior, with the outcome measured in a disjoint November-December window (a leakage-free design), separates returners from non-returners well: gradient boosting reaches ROC-AUC 0.823 ± 0.001 in five-fold cross-validation versus 0.739 for a recency-only rule and 0.500 for random ranking. The strongest signal associated with non-return is within-October recency; category breadth and number of active days are associated with returning. Behavioral clustering of all users into six descriptive personas shows the economics are highly concentrated: a modeled value exposure of \$572k (segment population $\times$ observed non-return rate $\times$ average October spend, a prioritization proxy - not realized lost revenue) sits 98% inside just two personas covering $\sim 10\%$ of users, while the highest-churn persona (91% non-return, 48% of all users) carries only $\sim \$3k$ of exposure.

Economic implication: Ranking segments by churn rate and by value exposure gives nearly opposite answers. A value-based targeting rule: Contact a customer only when *expected value* = churn risk $\times$ incremental uplift $\times$ customer value - contact cost is positive - selects 3.4% of holdout users with average October spend of \$81 versus \$2.56 for the rest, for an expected profit of \$6,382 on the 30% holdout under illustrative economics (\$2.70 per contact, 12% incremental uplift). A naive churn-first campaign contacting the entire highest-churn segment would spend $\approx \$301k$ per cycle against $\approx \$3k$ of exposure. All dollar outcomes are projections under stated assumptions, not realized savings.

Strategic recommendation: Shift retention budget from the highest-churn segment to the highest-exposure segments, trigger interventions from week-one behavior, and replace the originally proposed fixed 48-hour cart-recovery window with a tiered nudge (fast ~ 1 -hour primary, 24-48-hour fallback), because the observed median view-to-purchase lag is only ~ 19 minutes.

Experimental next step: None of the campaign economics have been validated causally - the data are observational. The immediate next step is the designed A/B test of the cart-recovery nudge: 3,216 users per arm detects a +3pp lift in the 60-day return rate (baseline 22.9%) at 80% power and $\alpha = 0.05$, with a fixed analysis date, sample-ratio-mismatch check, and guardrail metrics.

Business Context and Decision Framework - From a probability to a contact decision

The decision this project supports. Each month, the retention team must decide *which* recently active customers to contact with a retention offer, and *which* to leave alone. Contacting everyone is unaffordable; contacting by raw churn probability alone systematically selects low-value customers, because the customers most likely to leave are overwhelmingly those who never spent anything. The project therefore delivers (a) a calibrated churn-risk score, (b) descriptive customer segments, and (c) an expected-value decision rule that combines risk, customer value, an assumed incremental uplift, and contact cost.

Why churn probability alone is not sufficient. Probability answers “who is likely to leave?”; the business question is “where does a contact create positive expected value?” Those diverge for two reasons. First, value: a 91%-risk customer who spent \$0.03 in October is worth almost nothing to retain, while a 28%-risk customer who spent \$68 may clear the economic bar. Second, actionability: a retention offer only pays for the *incremental* customers it retains (the uplift), not for everyone contacted. Sections 12–14 build this logic out formally.

Element	Definition in this project
Population	All 232,079 users with ≥ 2 valid events in October 2019 after cleaning
Observation (feature) window	1–31 October 2019; all model inputs computed here
Outcome window	1 November – 31 December 2019 (~60 days), disjoint from the feature window
Outcome (“non-return”)	churn = 1 if the user records no events of any type in Nov–Dec; churn = 0 otherwise
Scoring time	At the feature-window cutoff (end of October), i.e., before any outcome data exists
Intervention	Retention contact (e.g., offer / reminder), assumed cost \$2.70 per contact
Primary outcome	60-day return rate (1 – non-return rate); baseline 22.9%
Decision rule	Contact user if $EV_i = p(\text{churn}_i) \times \text{incremental uplift} \times \text{customer value}_i - \text{contact cost} > 0$

Terminology used consistently throughout

Non-return: the observed outcome (no events in Nov–Dec). **Churn risk:** the model's calibrated probability of non-return. **Modeled value exposure:** the revenue-at-risk proxy (population $\times$ non-return rate $\times$ average spend). **Incremental uplift:** the assumed fraction of contacted would-be churners retained by the contact. **Customer value proxy:** total October purchase spends (not CLV).

Data and Cohort Construction - 20M raw events to a 232,079-user cohort

Source and granularity: The data are monthly event-log CSVs from a mid-sized cosmetics e-commerce store (the public REES46 “cosmetics shop” dataset), October 2019 through February 2020. One row is one logged user interaction with columns: event_time, event_type, product_id, category_id, category_code, brand, price, user_id, user_session. Valid event types are **view**, **cart**, **remove_from_cart**, **purchase**. October is the feature month; November-December define the outcome; January-February is used **only** for the revenue-trend context chart (§4.5) - no features or labels touch them.

Cleaning rules and why each one exists: (1) *Keep only the four valid event types*, anything else is instrumentation noise. (2) *Drop non-positive prices*, free/negative rows are not economic transactions and would distort spend features. (3) *De-duplicate* identical rows on (event_time, user_id, product_id, event_type), the feed logs some events twice; leaving duplicates would inflate every count feature. (4) *Brand back-fill*: missing brands are filled with the product's modal brand where any brand was ever logged for that product. (5) *Eligibility*: keep users with ≥ 2 events, since single-event rows carry no behavioral signal to model.

Cohort construction step	Effect (October)
Raw October events	$\approx 4.1\text{M}$ rows
→ valid event types + price > 0	non-economic rows removed
→ de-duplication	213,811 duplicate events removed
→ brand back-fill	only 127 of 1,565,034 missing brands recovered ($\sim 0.01\%$); the rest labelled “unknown”
→ users with ≥ 2 events	final cohort
Final modeling cohort	3,715,069 events · 232,079 users · 703,706 sessions · 1–31 Oct 2019

How cleaning decisions shape the target population: The ≥ 2 -events rule removes drive-by single-event visitors, so every reported rate (including the 77.1% non-return rate) describes users who showed at least minimal engagement; true sitewide non-return among all visitors is higher. The price filter means spend features measure economic activity only. Because category_code is $\sim 98\%$ missing on this feed, category breadth uses category_id (always present, 490 distinct values) — a deliberate substitution documented in the notebook.

Data-quality caveats carried through the analysis

Purchase events may not equal unique orders. The feed has no order identifier, so one checkout of five items appears as five purchase events. All “purchase count” features and the funnel's purchase stage count events, not orders.

Brand back-fill recovers almost nothing (127 of 1.57M missing values) because most products with a missing brand never have a brand logged; brand_breadth therefore mostly counts “unknown” as one brand. This is one reason the feature was later excluded from the model (§6).

Exploratory Analysis - A browse-heavy funnel with almost no habit formation

This section establishes the behavioral facts that motivate the label design (§5), the features (§6), and the recommendations (§15). Each result is presented as observation → interpretation → business implication → caveat. Four distinct quantities are kept separate throughout: event-count ratios (per-event volumes), user-level conversion rates (share of users reaching a funnel stage), within-October weekly retention (§4.6), and the 60-day non-return outcome (§5) - conflating these is a common source of error.

Conversion funnel - the cart → purchase step is the leak

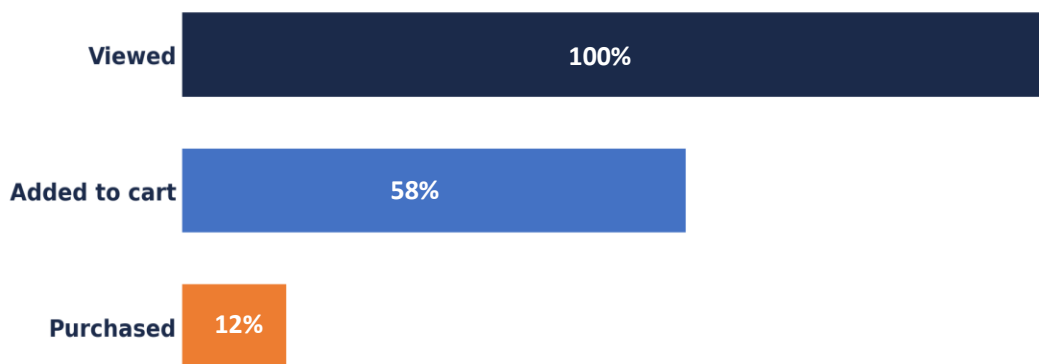


Figure 1. October event mix (left; per-event volumes) and user-level funnel (right; share of users reaching each stage). The two panels answer different questions and use different units.

Figure 1 shows two complementary views of customer activity during October 2019. The left panel summarizes the distribution of event volumes, while the right panel presents a user-level funnel indicating the proportion of customers who progressed through each stage of the purchase journey. Although 58.4% of users who viewed at least one product subsequently added an item to their cart, only 19.4% of those who reached the cart completed a purchase. Overall, just 25,755 customers (11.1% of the eligible cohort) made at least one purchase during the observation window. This pattern suggests that browsing and cart creation are common behaviours, whereas completing the checkout process represents the primary point of friction in

the customer journey. Consequently, the cart stage offers the greatest opportunity for retention interventions such as cart abandonment reminders or targeted incentives. The rapid purchase timing discussed in Section 4.3 further implies that any recovery campaign would need to be delivered shortly after cart creation to maximize its effectiveness. It is important to note that these figures describe within-October user behaviour only and should not be interpreted as evidence of future customer retention. In addition, the purchase stage is based on recorded purchase events rather than confirmed order records, meaning it should be interpreted as a behavioural proxy rather than an exact measure of completed transactions.

Engagement distributions - the median user has no habit

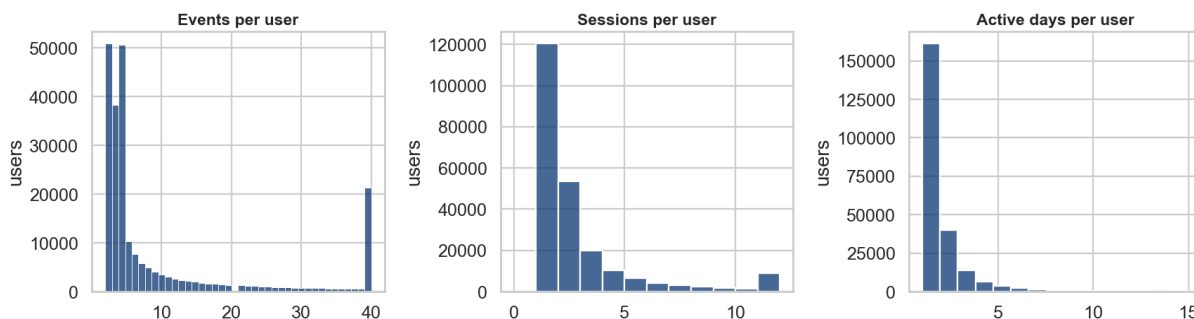


Figure 2. *Events, sessions and active days per user (clipped tails). All three distributions are heavily right-skewed.*

Figure 2 summarizes customer engagement through three complementary measures: total events, sessions, and active days per user. All three distributions are heavily right-skewed, with most customers generating only a small number of interactions and a long tail of highly active users. The majority of customers appeared in a single session on a single day, while relatively few returned across multiple sessions or days during October. This pattern suggests that the typical customer never developed a sustained shopping habit and repeated engagement represents the exception rather than the norm. From a business perspective, this finding highlights early engagement depth, particularly the number of sessions and active days, as potentially valuable indicators of future retention. Encouraging customers to return for a second session shortly after their initial visit may therefore represent an effective retention strategy. Because the distributions are strongly skewed, however, averages provide a poor summary of typical customer behavior. Throughout the remainder of the analysis, medians are used where appropriate, and the tree-based model is relied upon because it is naturally robust to highly skewed feature distributions.

Purchase timing - conversion happens within the hour

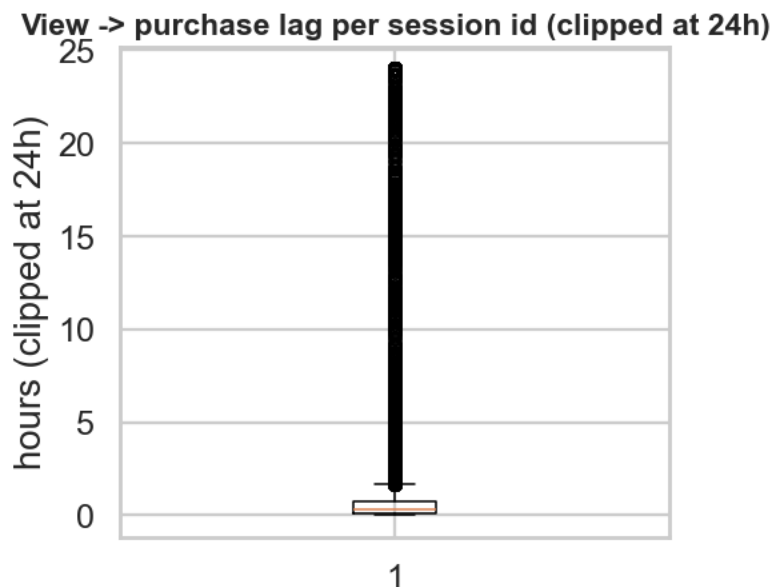


Figure 3. View→purchase lag per session id, clipped at 24h for display

Note: The join is per session id, an approximate (not guaranteed 1:1) match between a first view and a purchase.

The figure examines the time elapsed between a customer's first product view and a purchase within the same session, using 21,724 matched view-purchase session pairs. Although a small number of sessions exhibit very long delays, the distribution is highly concentrated near zero: the median view-to-purchase time is only 0.31 hours (approximately 19 minutes), and 75% of purchases occur within 45 minutes of the initial product view. The much larger mean (4.28 hours) reflects a long tail of delayed purchases extending to approximately 30 days, indicating that a small minority of customers deliberate substantially longer before completing a purchase. Overall, these results suggest that purchasing decisions in this category are primarily made during the initial shopping session rather than after extended consideration. Consequently, conventional cart-recovery campaigns triggered after 24-48 hours are likely to miss the period in which most conversions naturally occur. The evidence instead supports a tiered intervention strategy consisting of a rapid reminder—approximately one hour after cart creation—for the majority of customers, complemented by a later follow-up for the relatively small group of delayed purchasers. However, these timing patterns describe customers who ultimately converted and should not be interpreted as evidence that earlier reminders themselves increase conversion rates, establishing that causal relationship requires the randomized A/B experiment.

Price and spend - an impulse-priced category

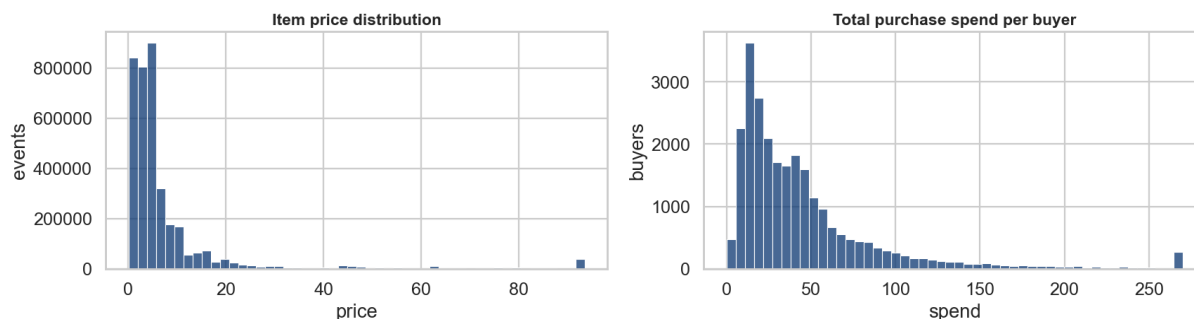


Figure 4. Item price distribution (left) and total October spend per buyer (right), clipped at the 99th percentile.

Next is the distribution of individual product prices alongside the total October spending of customers who completed at least one purchase. Product prices are concentrated at relatively low values, with a median item price of \$4.11, while the 25,755 buyers recorded a median monthly spend of \$32.38. These distributions indicate that the retailer operates primarily within an impulse-purchase category, where individual transactions are relatively inexpensive and customer value accumulates through repeated purchases rather than occasional high-value orders. This has important implications for retention strategy: Preserving long-term customer relationships is likely to generate greater economic benefit than optimizing the conversion of any single transaction. Accordingly, this study uses total October spending as a practical proxy for customer value when prioritizing retention interventions. Nevertheless, this measure should not be interpreted as customer lifetime value (CLV), since it excludes future purchases, gross margins, acquisition costs, and customer behavior beyond the observation window.

Revenue trend - Black Friday sits inside the outcome window

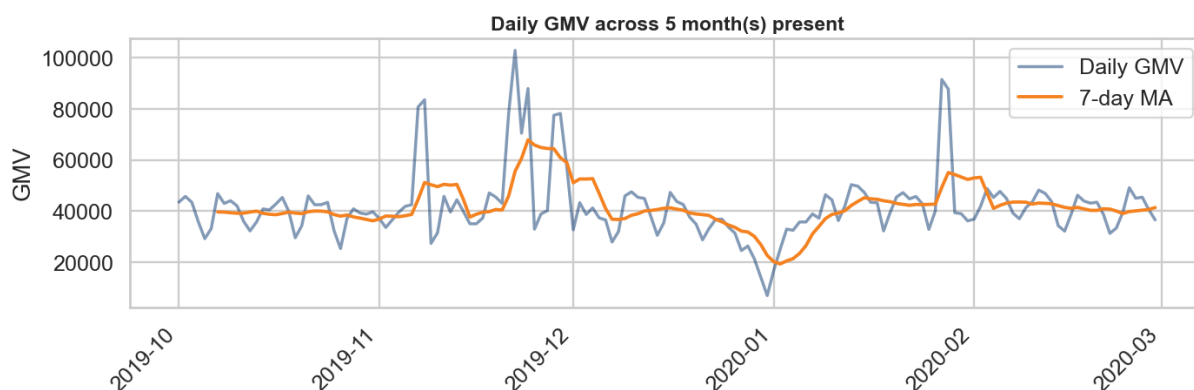


Figure 5. Daily GMV with a 7-day moving average across all five months present. GMV surges in November and dips in December as demand is pulled forward.

Revenue remains relatively stable throughout October before rising sharply during November, corresponding to the Black Friday promotional period, and subsequently declines during December as demand returns toward its earlier level. This pattern suggests that a substantial portion of November purchasing activity was promotion-driven rather than the result of organic customer loyalty. As a result, some customers classified as having "returned" during the 60-day

outcome window may have been motivated primarily by temporary promotional events rather than genuine long-term engagement. Consequently, the observed 77.1% non-return rate should be interpreted as a conservative estimate of underlying organic churn, and the predictive model may partially capture responsiveness to seasonal promotions in addition to persistent customer behavior. Importantly, although this figure includes data from January and February to provide broader business context, those months are excluded entirely from feature engineering and model labeling. Future work should evaluate model performance using an outcome window that is less affected by major promotional events.

Within-October cohort retention - decay is immediate

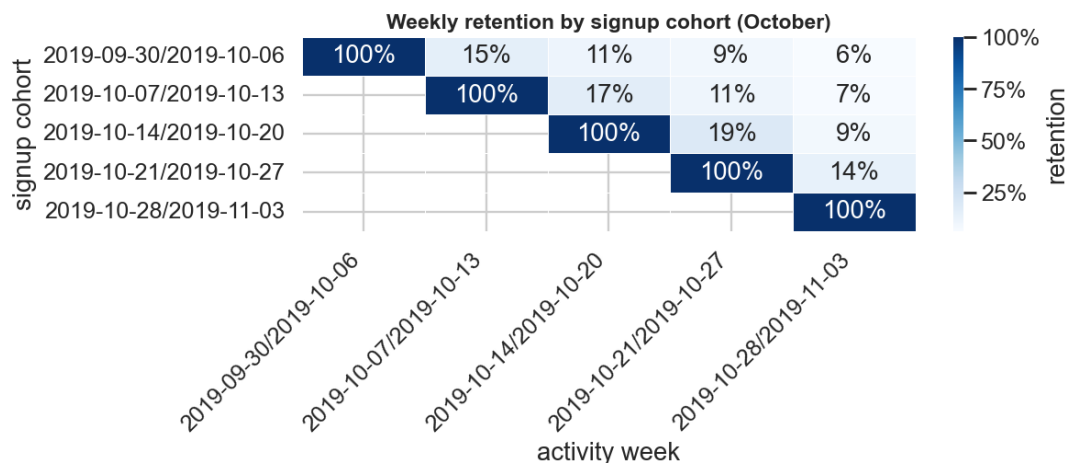


Figure 6. Weekly retention by first-active week within October. Most of each cohort never returns the following week.

Across every cohort, retention declines sharply after the initial week, with only a small minority of customers returning during the following week. This consistent pattern indicates that disengagement typically occurs very soon after a customer's first interaction rather than developing gradually over time. From a business perspective, the finding reinforces the importance of identifying at-risk customers based on their earliest behavioral signals. Waiting until prolonged inactivity becomes apparent would likely delay intervention until after many customers have already disengaged. It is important to distinguish this visualization from the churn outcome used throughout the predictive modeling stage. The heatmap measures weekly within-October activity, whereas the predictive model is trained on a separate 60-day non-return label constructed using future behavior after the observation window. The two measures therefore serve different analytical purposes and should not be compared directly.

Leakage-Free Outcome Design - Why the windows are disjoint

The failure mode this design prevents: A tempting single-month label is “churn = no activity in the last 30 days.” That label is mechanically the *same variable* as recency: feeding `days_since_last_active` into a model then “predicts” the label from a copy of itself, producing spectacular but meaningless metrics. Worse, with only one month of data such a label cannot even be observed. This is the canonical target-leakage error in churn modeling.

Features are computed **only from October**; the outcome is measured **only from November-December**. Because the windows share no time, every October feature, including within-October recency measured at the cutoff, is information genuinely available at scoring time, and none of it can encode the outcome.

Window	Dates	Role
Feature window	1 Oct - 31 Oct 2019	All 9 model features computed here (behavior, breadth, recency at cutoff, spend proxy)
Cutoff (scoring moment)	1 Nov 2019, 00:00	Recency = days from last October event to this instant; nothing after it is visible to the model
Outcome window	1 Nov - 31 Dec 2019	churn = 1 if the user records no events of any type in this window (~60 days)

days_since_last_active is measured *inside* October, against the cutoff. It correlates with the future outcome, that is what makes it predictive, but it is not derived from the outcome window. A leaky alternative would be any feature that peeks across the cutoff: a “returned-within-7-days” flag, recency computed relative to November activity, or November spend. Such features were identified and excluded by construction.

Two concrete label assignments:

Example	October behavior	Nov-Dec behavior	Label
Customer A	12 sessions across three weeks; two purchases; last event 30 Oct (recency 1 day)	One product view on 15 Nov	churn = 0 (returned - any event counts)
Customer B	One session on 2 Oct: four views, one cart, no purchase (recency 29 days)	No events at all	churn = 1 (non-return)

Resulting base rate: **77.1% of the cohort is labelled churn = 1** (non-return); the 60-day return rate is 22.9%. These two numbers anchor every baseline in §8 and the power analysis in §16.

Feature Engineering: Nine features across five behavioral signal groups

Each user's millions-of-rows event history is compressed into one row of interpretable behavioral summaries, organized in five groups: **activity intensity** (n_views, n_carts, avg_events_per_day), **habit / repeat engagement** (n_sessions, active_days), **product exploration** (cat_breadth), **purchase intent** (n_purchases, days_to_first_cart), and **recency** (days_since_last_active). **Customer value** (total October spend) is engineered but deliberately *kept out of the model* and reserved for the economic layer. The full dictionary:

Feature	Calculation (October only)	Behavioral meaning & expected direction	Missing values / caveat
n_views	count of view events	Browsing intensity; more views weakly associated with non-return (shallow browsing)	none
n_carts	count of cart events	Expressed intent; weakly associated with non-return in-model (cart-as-wishlist)	r = 0.70 with cat_breadth — flagged, retained
n_purchases	count of purchase events	Realized intent; expected protective	purchase events ≠ unique orders (no order id)
n_sessions	distinct user_session ids	Visit frequency (habit)	session id is feed-defined
cat_breadth	distinct category_id values	Exploration breadth; expected protective	uses category_id because category_code is ~98% missing
active_days	distinct calendar days with any event	Habit regularity; expected protective	none
avg_events_per_day	n_events / (last_seen – first_seen + 1 day)	Time-normalized intensity	+1 prevents division by zero for one-day users
days_to_first_cart	days from first_seen to first cart event	Speed of intent formation; expected protective (faster = stickier)	never-carted users have no value → filled with 31, i.e., “beyond the 31-day window” — an explicit ‘did not

Feature	Calculation (October only)	Behavioral meaning & expected direction	Missing values / caveat
			happen this month' encoding rather than a fake observed delay
days_since_last_active	days from last October event to the 1 Nov cutoff	Within-window recency; expected strongest risk signal	legitimate by design — see §5

Exclusions and collinearity

Two engineered features were **excluded after a correlation audit**, not for taste: **brand_breadth** ($r = 0.91$ with `cat_breadth`; adds no independent signal and mostly counts the “unknown” brand - §3) and **has_cart** ($r = -0.99$ with the filled `days_to_first_cart`; the two are near-duplicates). The exclusion had a diagnostic payoff: with `has_cart` present, the logistic coefficient on `days_to_first_cart` was an unstable -0.57 ; after removal it collapsed to -0.07 , confirming one signal had been split across two collinear columns. Among the nine retained features, one pair remains flagged at the audit threshold (`n_carts`–`cat_breadth`, $r = 0.70$), which is acceptable for the tree model, and reason coefficients are read jointly with permutation importance (§9).

Why total spend is excluded from the model but central to the business layer. Spend is the *value* term of the targeting rule (§13). Keeping it out of the risk model keeps the two axes of the decision, “how likely to leave” and “how much is at stake”, statistically separate and individually interpretable, and avoids the model partially re-learning value as risk.

Modeling Approach - Two baselines, one interpreter, one benchmark

Model	Purpose	Strength	Limitation
Majority-class baseline	Floor for every metric	Exposes prevalence-inflated metrics (PR-AUC = 0.771 with zero skill)	No ranking ability (ROC-AUC 0.500)
Recency-only rule	Honest single-signal benchmark	Shows how much of the problem one feature solves (ROC-AUC 0.739)	Ignores all other behavior
Logistic regression	Interpretable driver readout	Signed, standardized coefficients; fast; calibratable	Linear; sensitive to collinearity (audited in §6)
HistGradientBoosting	Predictive benchmark &	Captures non-linearities/interactions; robust to skewed counts	Opaque without post-hoc tools;

Model	Purpose	Strength	Limitation
	production candidate		needs calibration for probabilities

Training protocol (identical inputs for all models): 70/30 train–test split, stratified on the label, random_state = 42; identical split reused so every downstream number refers to the same 69,624-user holdout. Logistic regression runs in a pipeline with StandardScaler (so coefficients are per-standard-deviation and comparable) and class_weight="balanced"; max_iter = 1000. HistGradientBoosting uses max_depth = 4, learning_rate = 0.08, max_iter = 300, l2_regularization = 1.0, with balanced sample weights (returners up-weighted by the class ratio). days_to_first_cart is filled with 31 before fitting. Class imbalance (77/23) is thus handled by re-weighting rather than resampling, preserving the true base rate for calibration later.

Cross-validation. Five-fold stratified CV on the full cohort, with scaling and weighting fitted inside each fold (no preprocessing leakage). CV answers “is the single-split result stable?”, the single split answers “what exactly happens on one clean holdout that every later section can share?”

Validation and Model Performance - Read every metric against its baseline

Why PR-AUC must be read against prevalence? ROC-AUC has a fixed no-skill value of 0.500 regardless of class balance. PR-AUC's no-skill value equals the positive-class prevalence, which is **0.771**, because non-return is the *majority* class. A model that flags everyone as a churner already achieves precision 0.771 at every recall, so “PR-AUC 0.94” is only +0.17 above chance while “ROC-AUC 0.82” is +0.32 above chance. The honest headlines are therefore ROC-AUC and the lift over the recency-only baseline; PR-AUC is reported with its baseline printed beside it.

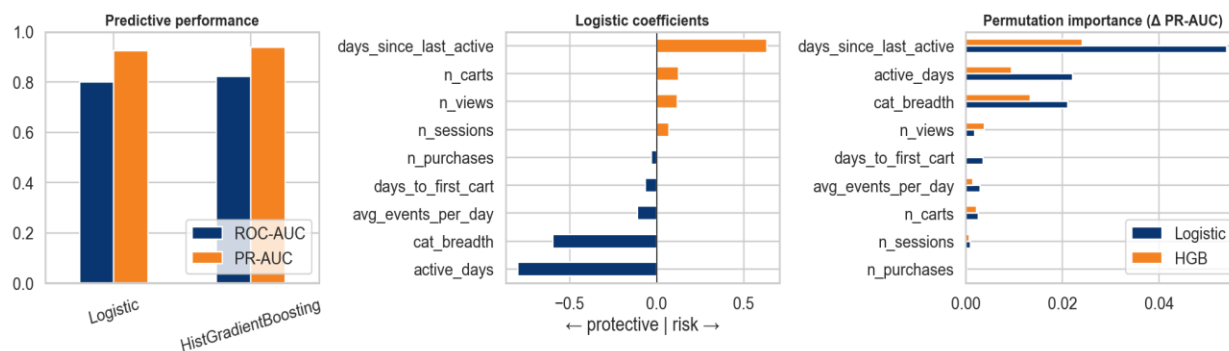


Figure 7. Holdout comparison: ROC and precision–recall curves against both baselines, and standardized logistic coefficients (right panel; direction of association).

Model	ROC-AUC (CV mean ± sd)	PR-AUC (CV mean ± sd)	Single- split holdout	Interpretability	Intended role
Majority class	0.500 ± 0.000	0.771 ± 0.000	—	trivial	metric floor
Recency only	0.739 ± 0.002	0.893 ± 0.001	—	total	single-signal benchmark
Logistic regression	0.801 ± 0.001	0.926 ± 0.001	0.801 / 0.926	high (signed coefficients)	driver readout
HistGradientBoosting	0.823 ± 0.001	0.939 ± 0.000	0.822 / 0.938	low (needs \$9 tools)	scoring model

Incremental lift, stated precisely: gradient boosting adds +0.323 ROC-AUC over random ranking, **+0.084 over recency alone**, and +0.022 over logistic regression. The first two gaps are material; the third is real but small - logistic regression recovers most of the achievable signal, which is itself informative: the drivers are largely additive. Fold-to-fold standard deviations of ≤ 0.002 across all models indicate the estimates are stable, and the tiny CV-vs-split differences (0.823 vs 0.822) are sampling noise, not disagreement.

Operating-point detail the headline hides

At the default 0.5 threshold the logistic model's precision on the minority “will return” class is only **0.440** (recall 0.689): when it predicts a user will come back, it is right less than half the time. Churners are flagged well (precision 0.889, recall 0.740; accuracy 0.728 on $n = 69,624$; HGB confusion matrix in Appendix A.5).

Positively identifying *loyal* users is the hard direction at this base rate, which is exactly why decisions in §13 use calibrated probabilities inside an economic rule rather than a raw 0.5 cutoff.

Brier score (mean squared error of the probability itself, lower is better) is 0.129 for the holdout scores - probability quality is assessed fully in §12, where it matters for the EV rule.

Model Interpretation - Recency dominates, breadth and habit protect

Two complementary lenses are used, and a driver is trusted only where they agree. **Standardized logistic coefficients** give the direction of association per standard deviation of a feature. **Permutation importance** (drop in PR-AUC when one feature's values are shuffled, 8-10 repeats with error bars) gives how much each model actually relies on the feature and works identically for both models. Coefficients can mislead under collinearity (a lesson demonstrated concretely - the $-0.57 \rightarrow -0.07$ coefficient collapse), permutation importance spreads credit across correlated features. Using both, all nine retained features pass the agreement test: consistent sign and positive importance in both models.

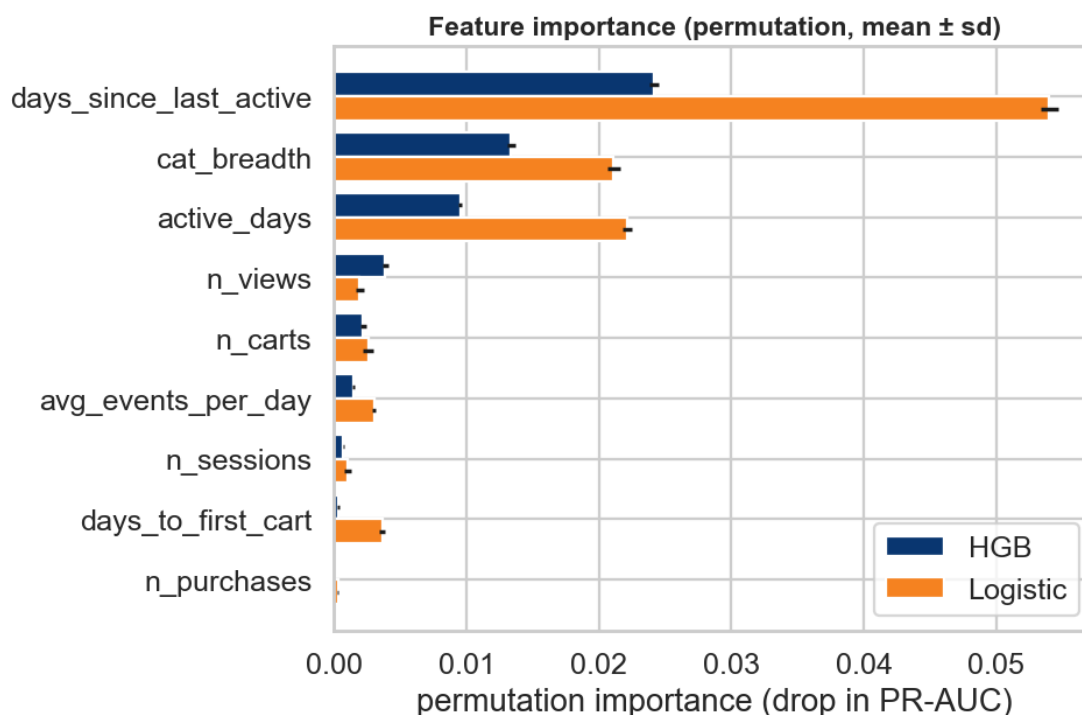


Figure 8. Permutation importance with fold-level error bars, both models. Recency dominates; category breadth and active days follow; everything else contributes small but consistent signal.

Driver	Direction	Logit coef (std.)	HGB perm. imp.	Business reading	Caveat
days_since_last_active	risk (+)	+0.633	0.0241	Every extra day of within-October silence is associated with higher non-return risk - by far the strongest signal	correlated with, not causing, disengagement
cat_breadth	protective (-)	-0.595	0.0134	Exploring more categories is linked to returning - breadth signals engagement with the store, not one product	$r = 0.70$ with n_carts ; read jointly
active_days	protective (-)	-0.796	0.0095	Spreading activity over more days is	

Driver	Direction	Logit coef (std.)	HGB perm. imp.	Business reading	Caveat
				predictive of returning - habit, not volume	
n_views / n_carts	risk (+)	+0.119 / +0.126	0.0039 / 0.0022	High raw volume without depth is weakly associated with non-return (window-shopping)	small effects; do not over-narrate
avg_events_per_day	protective (-)	-0.111	0.0015	Intensity per day is mildly protective	
n_sessions / days_to_first_cart / n_purchases	mixed, small	+0.071 / -0.067 / -0.031	≤ 0.0007	Consistent but minor contributions	n_purchases counts events, not orders

How to read this table: All statements are associations learned from observational data - “associated with”, “linked to”, “predictive of”. None establish that inducing breadth or extra active days would cause retention; that causal question is exactly what the experiment in §16 exists to answer. The pattern is nonetheless coherent and independently corroborated by the segmentation: the persona with strong purchase history but stale recency churns at 65%, while fresher, lighter users churn less (§10) - within-report evidence that recency dominates purchase history as a risk signal.

Customer Segmentation - Two views: buyers by value, everyone by behavior

Two segmentations answer two different questions on two different populations, and neither claims to discover “true” customer types - both are descriptive behavioral groupings produced by K-means, useful because they are stable enough to plan against, not because customers objectively belong to them. Cluster numbers are arbitrary identifiers with no inherent order.

RFM segmentation of the 25,755 buyers:

Method: Classic Recency-Frequency-Monetary features per buyer (recency to the 1 Nov cutoff, purchase-event count, total spend), transformed with log1p to tame heavy right skew, scaled to [0,1] with MinMaxScaler so no dimension dominates the distance metric, and clustered with K-means (k = 4 by elbow inspection; n_init = 10, seed 42, elbow chart in Appendix A.8).

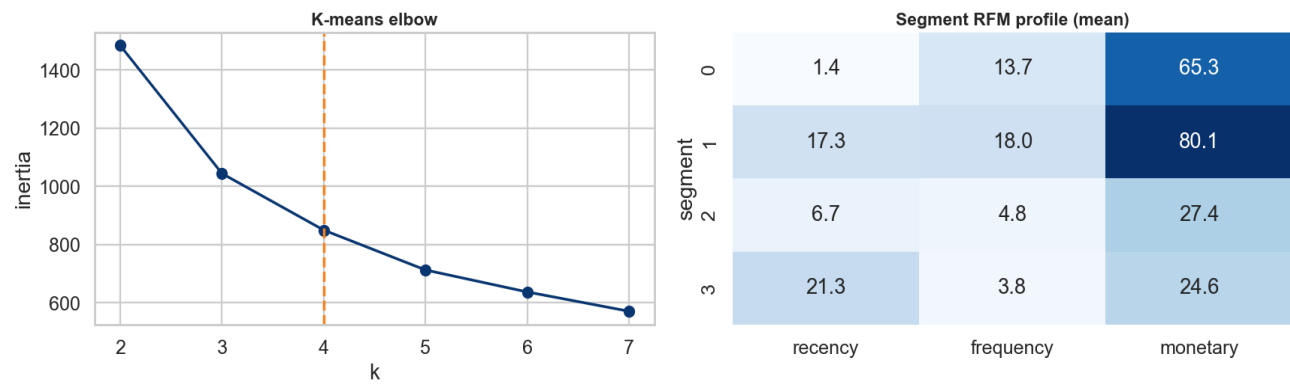


Figure 9. RFM segmentation of buyers: inertia elbow and segment structure. Churn rises monotonically as recency worsens and frequency/spend fall.

Segment (buyers only)	n	Recency (d)	Freq.	Monetary	Non-return	Strategic note
0 - Champions	3,268	1.4	13.7	\$65.3	37.9%	Most engaged buyers; even here 1 in 3 does not return
1 - High-value, slipping	7,743	17.3	18.0	\$80.1	42.1%	Highest frequency & spend, but a 17-day recency gap - the costliest group to lose, flag first
2 - Mid-value, recent	5,215	6.7	4.8	\$27.4	56.2%	Growable middle
3 - One-time / low-value	9,529	21.3	3.8	\$24.6	65.7%	Largest buyer tier, lowest expected payoff for retention spend

The limitation that motivates the second view: RFM sees only the 11.1% of the cohort that purchased. The other ~89%, where most non-return happens, is invisible to it.

Behavioral personas for all 232,079 users:

Method: Six features computed strictly on the October window for every user - recency_days, purchase_cnt, monetary, view_cnt, cart_cnt, remove_cnt - log1p-transformed, scaled, and clustered with K-means (k = 6, chosen from the elbow plus the need to separate browsers / carters / researchers / efficient buyers; seed 42). Names below are editorial labels attached to cluster medians, not discovered categories.

Persona (all users)	Size (share)	Recency (d)	Purch. (med.)	Spend (med.)	View→Purch.	Non-return
5 - Champions	8,497 (3.7%)	2	11	\$46.6	30.7%	27.8%
1 - Efficient repeat buyers	15,888 (6.8%)	16	5	\$27.6	53.7%	65.0%
3 - Cart hoarders / researchers	14,930 (6.4%)	9	0	\$0	0.0%	50.1%
4 - Fresh light browsers	27,282 (11.8%)	1	0	\$0	0.7%	58.1%
0 - Passive one-and-done	54,143 (23.3%)	9	0	\$0	0.1%	76.5%
2 - Stale / disengaged	111,339 (48.0%)	25	0	\$0	0.2%	91.1%

Reading the personas strategically: *Champions* (3.7%) are the revenue backbone: freshest recency, highest purchase depth, best cart follow-through. *Efficient repeat buyers* convert more cleanly than anyone when they engage (53.7% view→purchase), yet churn at 65%, and the only profile difference is stale recency (16 days vs Champions' 2): the segmentation independently confirms §9's headline that recency dominates purchase history. *Cart hoarders* add heavily but never buy - a pure “add and hesitate” group already at the top of the funnel, the natural audience for cart recovery. *Fresh light browsers* are too new to classify. *Passive one-and-done* sits at the base rate. *Stale / disengaged* (48% of all users) combines the worst recency with essentially zero conversion and the highest non-return (91.1%); RFM never sees this group at all.

Persona data-quality note

The stale cluster's median view count (1) is *below* its median cart count (2): some cart events arrive without a matching logged view (e.g., direct navigation or tracking gaps). A data-quality observation, not a behavioral one, and a further reason event counts should not be read as precise journey steps.

Modeled Value Exposure - 98% of the exposure sits in two personas

Definition:

Modeled value exposure (revenue-at-risk proxy)

exposure(segment) = segment population × observed non-return rate × average observed October spend

This is a **prioritization proxy**: it ranks where value and risk co-locate. It is **not** realized lost revenue (nothing has been lost yet), **not** a customer-lifetime-value estimate (October spend only, no margin), and **not** a causal forecast (it does not say a campaign could retain it). Its single legitimate use is deciding which segments deserve attention first.

Persona	n	Non-return	Mean spend	Expected users lost	Modeled value exposure	Share
1 — Efficient repeat buyers	15,888	65.0%	\$38.68	10,335	\$399,758	69.9%
5 — Champions	8,497	27.8%	\$68.32	2,366	\$161,645	28.3%
4 — Fresh light browsers	27,282	58.1%	\$0.40	15,858	\$6,343	1.1%
2 — Stale / disengaged	111,339	91.1%	\$0.03	101,420	\$3,043	0.5%
0 — Passive one-and-done	54,143	76.5%	\$0.03	41,433	\$1,243	0.2%
3 — Cart hoarders	14,930	50.1%	\$0.02	7,480	\$150	0.0%
Total	232,079	77.1%	—	—	\$572,182	100%

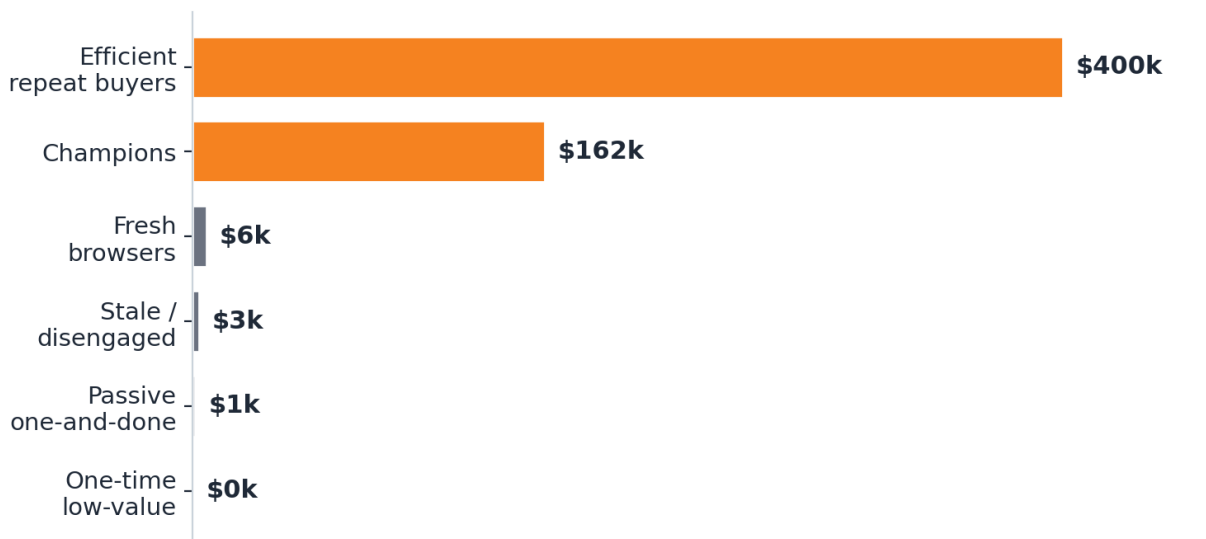
Revenue at risk by customer segment (60-day horizon)

Figure 10. Modeled value exposure by persona over the 60-day horizon. Concentration, not churn rate, is the story: \$561k of \$572k sits in two segments covering 10.5% of users.

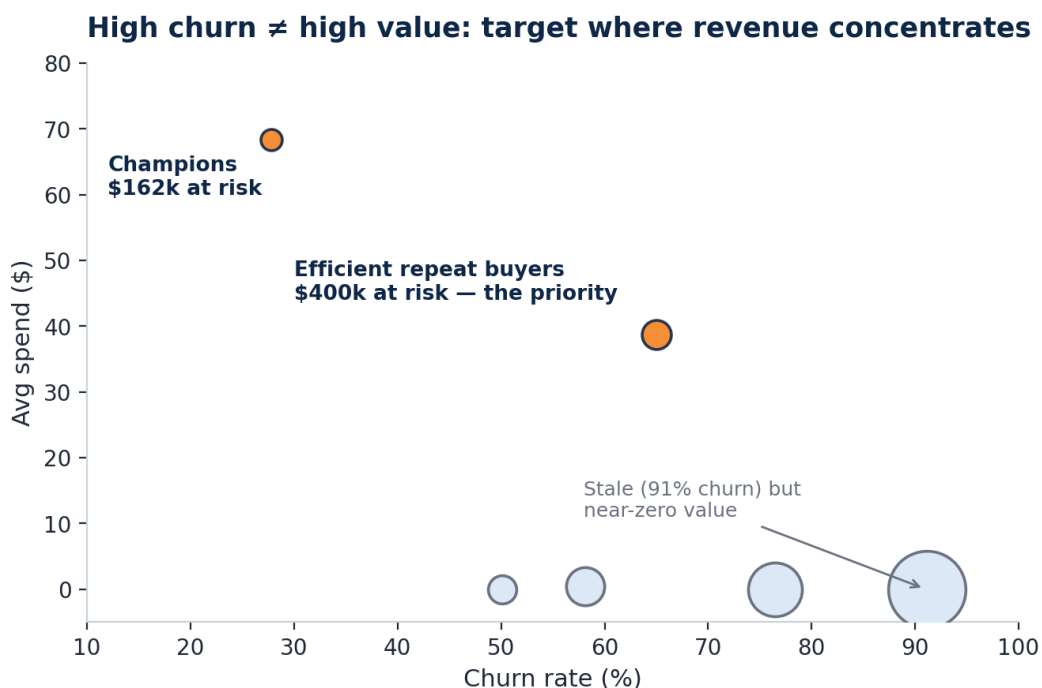


Figure 11. Non-return rate versus average spend (bubble size = persona population). The highest-churn segment and the highest-exposure segments are different segments.

Why a high-churn segment can be economically unattractive? The stale segment's 91% non-return looks like the emergency, but its members average \$0.03 of October spend: even a costless, perfectly effective campaign could only protect ~\$3k. Contact costs make it strictly value-destroying to pursue at high touch (quantified in §14). Conversely, Champions churn least (27.8%) but rank second in exposure purely through spend, therefore indicating that lower-risk, high-value customers can be worthwhile targets. This inversion is the central strategic finding of the report, and it is why the targeting rule in §13 multiplies risk by value rather than ranking on risk alone. Note the mean spends in this table (\$38.68 / \$68.32) exceed the persona medians in §10 (\$27.6 / \$46.6), spend is right-skewed, and the exposure calculation deliberately uses the mean because totals, not typical users, are what a budget protects.

Probability Calibration - The scores can be trusted as probabilities

Ranking and probability accuracy are different properties: ROC-AUC certifies only that churners tend to score above non-churners; it is invariant to any monotone distortion of the scores. The EV rule in §13, however, multiplies the score by dollars - “ $p = 0.8$ ” must actually mean “~80 of 100 such users fail to return,” or every expected-value computation inherits the distortion. Calibration is what licenses that reading.

Method: Isotonic regression via CalibratedClassifierCV with 5-fold internal cross-validation wrapped around the HGB model, fitted on training data only and evaluated on the untouched 30% holdout. Quality is read from the **Brier score** (mean squared error between predicted probability and 0/1 outcome) and the **reliability curve** (predicted vs empirical frequency by bin).

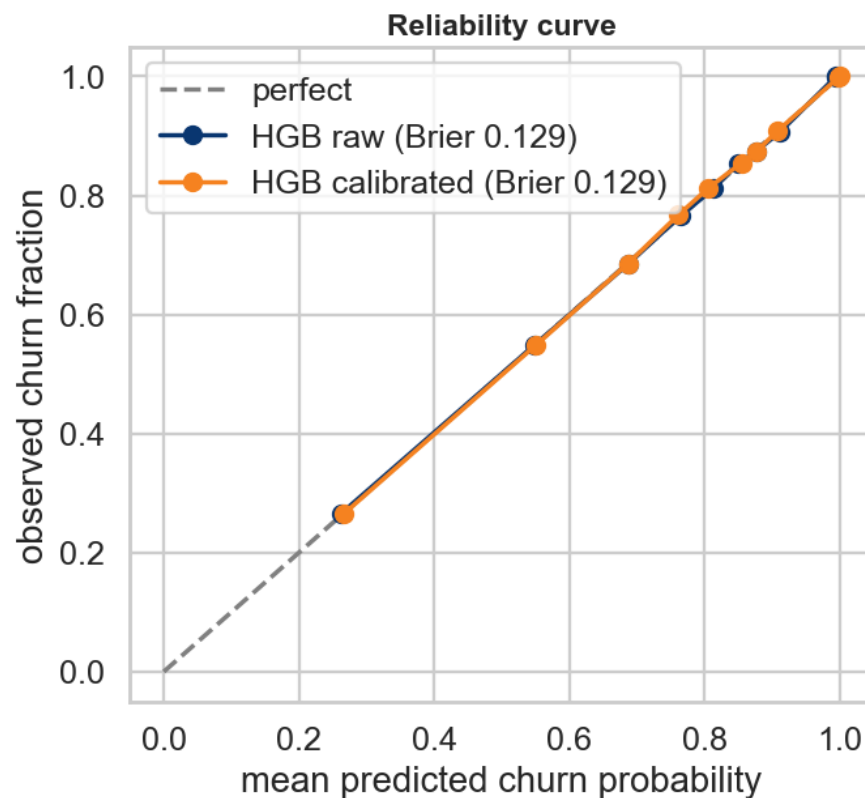


Figure 12. Reliability curve and score distribution on the holdout. The raw model already tracks the diagonal closely; isotonic calibration changes little.

Result: Brier 0.129 raw vs 0.129 calibrated - identical to three decimals. Calibration did not materially improve the model; its value here is validation: it demonstrates the balanced-weighted HGB's raw probabilities were already trustworthy, so the EV arithmetic in §13 stands on measured probability quality rather than hope. Holdout score distribution: mean 0.770 (matching the 0.771 prevalence, as a calibrated model should), median 0.829.

Known artefact: probability pile-up at the top

Isotonic calibration maps **14.8% of holdout users to scores ≥ 0.999** (21.1% score ≥ 0.99). Within that top mass, ranking resolution is essentially gone - the model can say “virtually certain non-returner” but cannot finely order users inside that group.

Consequence for targeting: among the highest-risk users, prioritization is driven almost entirely by the value term of the EV rule, not by risk differences. This is acceptable for §13's purpose and disclosed here rather than hidden.

Value-Based Targeting - Contact where risk $\times$ uplift $\times$ value clears cost

The decision rule

$EV_i = p(\text{churn}_i) \times \text{incremental uplift} \times \text{customer value}_i - \text{contact cost}$, contact if $EV_i > 0$.

$p(\text{churn}_i)$: calibrated non-return probability (§12). **Incremental uplift**: fraction of contacted would-be churners the campaign retains; set to **0.12**. **Customer value_i**: October spend proxy. **Contact cost**: **\$2.70** per higher-touch contact (Assumed).

Equivalently: contact if value-at-risk $p_i \times \text{value}_i > \text{cost} / \text{uplift} = 2.70 / 0.12 = \mathbf{\$22.50}$ (break-even). Omitting uplift from the formula - a simplification that appeared in an earlier draft of this report - understates the break-even by $\sim 8\times$ and would flag wildly unprofitable contacts, the uplift term is not optional, because a campaign is paid for only by the incremental retentions it causes.

Why each qualitative behavior of the rule is right? Churn probability alone is insufficient because it ignores stakes (§11). High-risk, low-value users fail the rule: $p = 0.91 \times 0.12 \times \0.03 is microscopic against \$2.70. Lower-risk, high-value users can pass it: $p = 0.30 \times 0.12 \times \$80 = \$2.88 > \2.70 . The rule thus automatically reproduces (per individual user, not per segment) the strategic inversion of §11.

Holdout evaluation (n = 69,624) - the number that counts as evidence

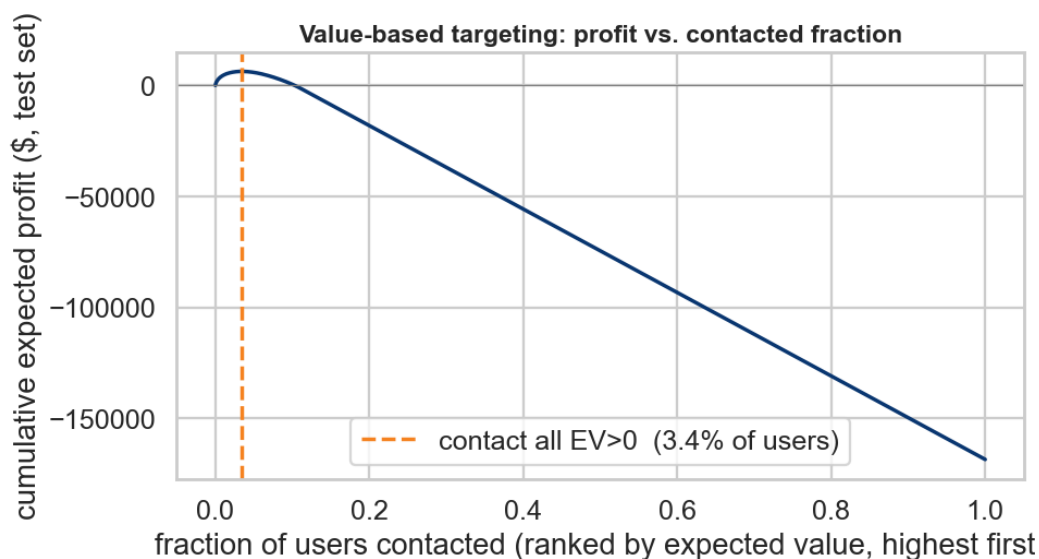


Figure 13. EV-positive selection on the holdout: who is targeted, and the spend gap between targeted and non-targeted users.

Applying the rule to the 30% holdout selects **2,388 users (3.4%)**. Their mean October spend is **\$81.09 vs \$2.56** for the untargeted; **60.1% of them actually churned** - below the 77.1% base rate by design, because prioritizing value favors higher-spend segments that churn less (this is a feature of the rule, not a defect, and it is a share-that-churned, **not** campaign lift). Summed EV of the selection: **expected profit \$6,382 under the stated assumptions**.

Budget-constrained top-K and why the fixed budget loses money

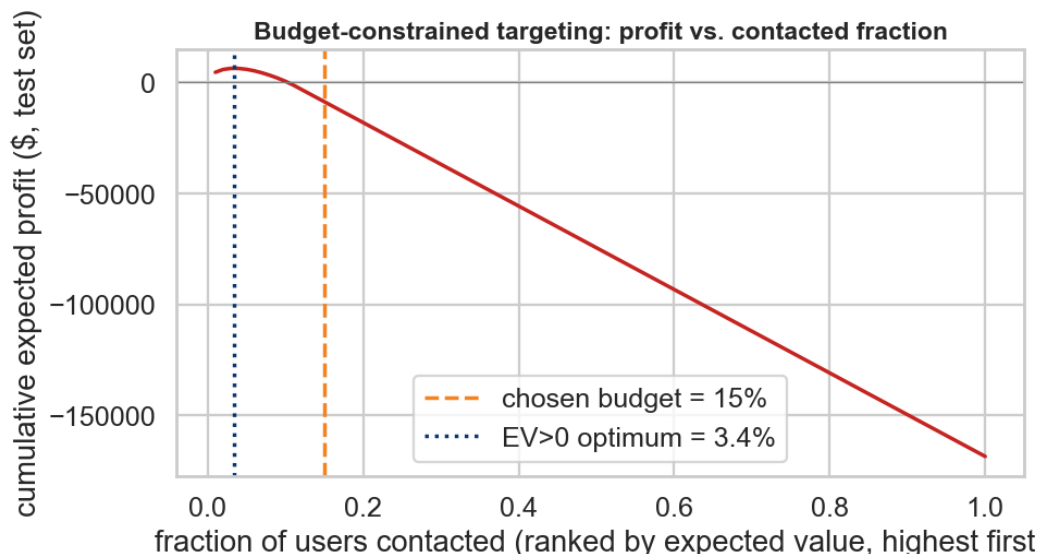


Figure 14. Expected profit as the contacted fraction grows (users ordered by EV). Profit peaks near the EV-positive share and turns negative well before 15%.

A fixed “top 15% by EV” budget (10,443 users) yields **expected profit –\$8,651**: past the ~3.4% of genuinely EV-positive users, every additional contact has negative expected value by construction, and 15% forces ~8,000 such contacts. Practical rule: **spend down the EV curve and stop at break-even; do not spend to a round budget number.**

Deployment-style scoring of the full base: Application, not validation

Scoring all 232,079 users with the trained model and applying the same rule targets 3.4% of the full base, so $\approx 7,900$ users with this persona composition: **37.4% of Efficient repeat buyers, 21.6% of Champions, 0.2% of Fresh light browsers, and 0.0% of every other persona**, the rule's per-user arithmetic lands almost exclusively on the two segments holding 98% of the modeled value exposure, with zero manual segment selection.

Scope discipline

The **holdout** numbers (2,388 targeted; \$6,382) are evaluation on unseen users and constitute the report's evidence. The **full-base** numbers ($\approx 7,900$ targeted) are a deployment-style application that includes users the model was trained on, they describe what the campaign list would look like and must not be quoted as model validation.

Financial Scenario Analysis - Every dollar figure labelled and stress-tested

Assumptions register

Input	Value	Basis / sensitivity
Contact cost	\$2.70 per user	Illustrative higher-touch outreach (SMS/call + offer handling), no cost data in the dataset
Incremental uplift	12% of contacted would-be churners retained	Illustrative, retention-campaign uplifts vary widely, must be measured by the \$16 experiment
Customer value proxy	Total October spend	Not CLV: excludes margin, future months, acquisition cost; revenue $\neq$ gross margin
Budget share (alt. rule)	Top 15% by EV	Tested and rejected (\$13.2)
Offer cost beyond contact	Not modeled	Any discount redeemed adds cost per retained user, treat current profit figures as upper bounds
Non-return rate / spends	As measured (\$5, \$11)	Empirical, but Nov-Dec window includes Black Friday (\$4.5)

Strategy comparison - One campaign cycle

	Churn-first (naive)	Value-first (recommended)
Selection rule	Contact the entire highest-churn persona (stale / disengaged)	Contact if EV > 0
Users contacted / cycle	111,339	$\approx$ 7,900 full-base (3.4%); 2,388 on holdout
Campaign spend / cycle	$\approx$ \$300,600 (7,900 $\times$ \$2.70)	$\approx$ \$21,300 full-base
Modeled value exposure addressed	$\approx$ \$3,043 (0.5% of total)	\$561k in the two personas the selection concentrates in (98%)
Spend per \$1 of exposure addressed	$\approx$ \$98.8	$\approx$ \$0.04-0.05
Expected profit / cycle	Deeply negative under any plausible uplift	\$6,382 on holdout; $\approx$ \$21k if scaled linearly to the full base

“Spend per \$1 of exposure addressed” is a prioritization-efficiency ratio, it compares campaign cost to the exposure pool the campaign points at. It does **not** mean \$1 of revenue is defended per \$0.05, because only uplift $\times$ exposure could ever be retained, and uplift is assumed. Similarly, the $\approx$ \$21k full-base profit divides the holdout result by the 30% holdout share, real profit depends on uplift, cost, and offer redemption, none of which this dataset measures.

Sensitivity - how the break-even moves

Because users are selected by $p \times \text{value} > \text{cost} / \text{uplift}$, the assumption pair (cost, uplift) sets who is contacted through a single number - the break-even value-at-risk (cost / uplift). Re-running the holdout selection across three cost and three uplift levels gives the full grid below, all outcomes are on the 30% holdout, on the same footing as the \$6,382 base case.

Contact cost	Uplift	Break-even $p \times$ value	Targeted (share)	Churn% of targeted	Expected profit (holdout)
\$1.35 (low)	6%	\$22.50	2,388 (3.4%)	60.1%	\$3,191
\$1.35 (low)	12%	\$11.25	4,627 (6.6%)	58.5%	\$10,925
\$1.35 (low)	24%	\$5.63	6,527 (9.4%)	57.4%	\$29,351
\$2.70 (base)	6%	\$45.00	720 (1.0%)	61.4%	\$1,353
\$2.70 (base)	12%	\$22.50	2,388 (3.4%)	60.1%	\$6,382 ← base
\$2.70 (base)	24%	\$11.25	4,627 (6.6%)	58.5%	\$21,850
\$5.40 (high)	6%	\$90.00	153 (0.2%)	66.0%	\$431
\$5.40 (high)	12%	\$45.00	720 (1.0%)	61.4%	\$2,706
\$5.40 (high)	24%	\$22.50	2,388 (3.4%)	60.1%	\$12,764

**All profit and share figures above are holdout projections under the stated assumptions*

Implications:

- Assumptions dominate the economics: Expected holdout profit spans \$431 to \$29,351 across the nine corners - a $\sim 68\times$ range driven entirely by two inputs the dataset does not measure, which is the quantitative case for gating any budget on the *measured* uplift from \$16 rather than the assumed 12%.

- Break-even fixes who to target, magnitude scales separately: The three corners that share a \$22.50 break-even - (\$1.35, 6%), (\$2.70, 12%), (\$5.40, 24%) - select the identical 2,388 users (3.4%, 60.1% churn), yet return \$3,191 / \$6,382 / \$12,764, because profit also scales with the absolute cost/uplift level, not just their ratio. Doubling the bar to \$45 (halving uplift or doubling cost) shrinks the list to $\sim 1.0\%$ and profit toward \$1-3k; the low-cost/high-uplift corner (\$5.63 bar) roughly triples the pool to 6,527 users and lifts holdout profit to $\sim$ \$29k.

- Value-based selection is robust to the assumptions: Churn among the targeted stays in a narrow 57-66% band at every corner so prioritizing value keeps favoring lower-churn, higher-value users no matter where cost and uplift land. Two structural conclusions also survive every corner: the stale segment never becomes economic at high touch (its value-at-risk is capped near $\$0.03 \times 0.91$), and the two high-exposure personas dominate the selection under all nine scenarios.

Strategic Recommendations - Four moves, in priority order

The preceding analyses identify four intervention opportunities that differ in expected value, implementation complexity, and supporting evidence. Recommendations are therefore prioritized according to their estimated business impact rather than predictive confidence alone.

R1. Tiered Cart-Recovery Nudge (Highest Priority)

Target customers: Customers who added at least one item to their cart but did not complete a purchase. The primary reminder should be triggered approximately one hour after cart abandonment, followed by a secondary reminder 24-48 hours later for customers who remain inactive.

Business rationale: The purchase timing analysis shows that purchasing decisions occur very quickly on this platform, with a median view-to-purchase lag of only 19 minutes and three-quarters of conversions completed within 45 minutes. This suggests that customer intent decays rapidly after a cart is created, making conventional 24-48 hour cart recovery campaigns too late for most users. A tiered reminder strategy aligns the intervention with observed customer behavior by reaching the majority of customers while purchase intent is still active, while retaining a later reminder for the relatively small group of delayed purchasers.

Success metrics: Cart-to-purchase conversion rate; 60-day return rate among contacted customers.

Key risks: Frequent reminders may increase notification fatigue, while promotional incentives could cannibalize purchases that would have occurred without intervention.

Validation required: A randomized A/B experiment should compare the proposed tiered reminder schedule against the existing recovery strategy to quantify incremental conversion and long-term retention.

R2. High-Value Retention Program

Target customers: Customers belonging to the Efficient Repeat Buyers and Champions personas whose expected value (EV) exceeds zero, representing approximately 37% and 22% of those respective segments.

Business rationale: The value-exposure analysis identifies these two personas as containing the overwhelming majority of economically meaningful retention opportunities. Rather than allocating budget solely according to churn probability, marketing resources should prioritize customers who combine elevated churn risk with meaningful customer value. Personalized replenishment reminders, loyalty incentives, and VIP return offers are therefore expected to deliver substantially greater financial returns than untargeted retention campaigns.

Success metrics: 60-day non-return rate within the high-value population; Incremental profit per contacted customer.

Key risks: The current business case assumes a 12% treatment uplift and a fixed campaign cost. If these assumptions differ materially from operational reality, the estimated financial return may change.

Validation required: Conduct an A/B experiment within the EV-selected customer list and replace the assumed uplift with the experimentally observed treatment effect in future targeting models.

R3. Early-Activation Onboarding

Target customers: New customers during their first week of activity, particularly those who interact with the platform only once, including the Fresh Light Browsers and Passive One-and-Done personas.

Business rationale: Exploratory analysis demonstrates that most customers never progress beyond a single session or a single active day. Furthermore, the predictive models consistently identify repeated sessions and broader category exploration as protective behavioral signals associated with future return. An onboarding experience that encourages customers to revisit the platform, explore additional categories, or complete an initial purchase may therefore strengthen early engagement before disengagement becomes permanent.

Success metrics: Sessions per user during the first week; Week-one to week-two retention rate.

Key risks: The observed relationships are associative rather than causal. Increasing browsing activity or category exploration does not necessarily guarantee higher retention.

Validation required: Evaluate alternative onboarding experiences through a randomized A/B experiment and measure their impact on repeat engagement and long-term retention.

R4. Low-Value Suppression

Target customers: Customers classified within the Stale / Disengaged persona, representing approximately 48% of the eligible customer base while generating an average observed October spend of only \$0.03.

Business rationale: Financial analysis indicates that broad retention campaigns targeting this segment are unlikely to recover sufficient customer value to justify their cost. High-touch interventions should therefore be avoided, with communication limited to zero- or near-zero-cost channels such as automated reminder emails. Importantly, suppression should be governed by the expected value framework rather than persona membership alone, ensuring that any customers with positive expected value continue to receive appropriate interventions.

Success metrics: Campaign cost per marketing cycle; Gross merchandise value (GMV) within the suppressed population as a guardrail metric.

Key risks: Although the segment is economically weak on average, a small number of valuable customers may still be present. Blanket exclusion based solely on segmentation could therefore eliminate profitable retention opportunities.

Validation required: Compare suppressed and non-suppressed customer groups during the initial deployment phase to confirm that marketing costs decline without producing a measurable loss in revenue.

#	Value	Basis
1	Tiered Cart-Recovery Nudge	Recover abandoned carts while purchase intent remains high
2	High-Value Retention Program	Protect customers with the greatest expected economic value
3	Early-Activation Onboarding	Build repeat engagement during the customer's first week
4	Low-Value Suppression	Reduce marketing spend on economically unattractive customers

A/B Testing Plan - Designed, powered, and honest about what is simulated

Eligibility & randomization: Unit: user (avoids within-user contamination across sessions). Eligible: new users who abandon a cart during the enrolment window. Assignment: 50/50 at first qualifying abandonment, held fixed. **Treatment:** tiered nudge (R1). **Control:** no nudge. **Primary metric:** 60-day return rate - chosen over immediate conversion because retention, not the single checkout, is the economic objective. **Secondary:** cart→purchase conversion, revenue per user. **Guardrails:** notification opt-out rate and revenue per user must not degrade.

Design element	Value	Basis
Baseline return rate p_0	22.9%	$1 - 0.771$
Minimum detectable effect	+3.0pp absolute (→ 25.9%)	Smallest lift worth operationalizing

Design element	Value	Basis
Significance / power	$\alpha = 0.05$ (two-sided), $1 - \beta = 0.80$	Convention
Sample size	3,216 per arm; 6,432 total	Two-proportion formula, Appendix A.11
Duration	~2 days at ~5,000 eligible users/day	Traffic; realistically run 1–2 weeks to cover weekday/weekend cycles
Analysis	Two-proportion z-test; report the lift with its 95% CI, not a bare p-value	Pre-registered, fixed analysis date
Integrity checks	Sample-ratio-mismatch χ^2 on arm sizes; intention-to-treat (analyze as assigned, whether or not the nudge was seen)	Protects against allocation bugs and selection effects

Simulated readout: With an injected true lift of +5pp, the simulated test returns control 23.3% vs treatment 26.1%, lift +2.8pp (95% CI +0.7 to +4.9pp), $z = 2.63$, $p = 0.0085$, SRM $p = 1.000$. This demonstrates the mechanics and readout format only, it is a random draw from an assumed effect and is evidence of nothing about the real campaign.

Peeking penalty: Checking once at the fixed date yields a 4.8% false-positive rate (nominal 5%); peeking five times and stopping early inflates it to **13.7%**. Hence: fix N and the analysis date in advance or adopt formal sequential testing / alpha-spending. A CUPED adjustment using October engagement as covariate is the recommended variance-reduction upgrade to shrink the required sample.

Implementation Roadmap - Prove, scale, institutionalize

Phase	Timing	Deliverables	Owner	Exit / success criterion
1 - Prove	Weeks 1-4	Validate cost & uplift assumptions with marketing finance; nightly scoring table (calibrated risk $\times$ spend proxy) into CRM; CRM eligibility & suppression lists (R4); launch R1 A/B per \$16; baseline dashboards	DS + CRM ops	A/B live with SRM clean; risk score consumed by CRM; assumptions register signed off
2 - Scale	Months 2-3	A/B readout (lift + CI); roll out winning nudge variant; launch R3 onboarding test; wire EV > 0	DS + lifecycle marketing	$\geq +3pp$ 60-day return at 95% confidence on

Phase	Timing	Deliverables	Owner	Exit / success criterion
		selection into campaign tooling with the measured uplift replacing the 12% assumption		R1; +5pp cart→purchase; EV targeting automated
3 - Institutionalize	Months 4-6	Monthly retrain with drift monitoring; treatment-history logging; first uplift model (targets <i>persuadability</i> , not just churners); upgrade value proxy toward margin/CLV; out-of-time validation on a non-promotional window; quasi-experimental read (PSM / DiD) on historical data	DS platform	Out-of-time ROC-AUC $\geq$ 0.80; profit per cycle sustained at the measured-uplift projection; uplift model beats EV rule in test

** Phase 1 converts assumptions into measurements; Phase 2 spends only on what measured; Phase 3 replaces the two structural shortcuts this report is explicit about - churn probability as a stand-in for persuadability, and October revenue as a stand-in for customer value.*

Limitations and Risks - Each with its business consequence

Limitation	Business consequence
October-only feature window. One month of behavior per user.	Seasonal shoppers and slow-cadence buyers are mis-profiled; scores for low-activity users rest on thin evidence. Extend feature windows in Phase 3.
Nov–Dec outcome window overlaps Black Friday.	Some “returns” are deal-chasing, so 77.1% non-return is a lower bound on organic churn and the model partly learns promo-responsiveness; out-of-time validation on a clean window is required before trusting scores year-round.
Non-return $\neq$ permanent churn.	A customer buying again in February counts as churned here. Campaign sizing based on this label overstates truly lost customers; treat it as a re-engagement trigger, not a death certificate.
Purchase events $\neq$ unique orders (no order id).	Frequency features and funnel purchase counts are inflated for multi-item checkouts; cross-store comparisons of these figures are invalid.
October spend is a value proxy, not CLV; revenue is not gross margin.	EV targeting may over-prioritized high-revenue/low-margin buyers and under-prioritized growing

Limitation	Business consequence
	customers; profit figures are upper bounds before margin data.
No promotion, pricing, or acquisition-channel data.	Recommendations are product/UX levers only; nothing here attributes churn to campaigns or channels, and channel-level ROI cannot be computed.
Behavioral associations are not causal.	Boosting breadth or sessions may not move retention at all; every recommendation is gated behind an experiment for exactly this reason.
Segmentation stability untested over time.	Persona sizes/boundaries may drift month to month; a campaign hard-wired to October's clusters can silently mis-target by December. Re-fit and monitor in Phase 3.
Temporal model stability unproven.	All validation is within-period; a holiday-season model may degrade in spring. Out-of-time validation is a Phase 3 exit criterion.
Uplift (12%) and contact cost (\$2.70) are assumptions.	Every dollar projection moves near-linearly with them (\$14.3); no budget commitment should precede the Phase 1 measurement.

Conclusion

On a browse-heavy, low-ticket cosmetics store, 77.1% of October-active users do not return within 60 days; a leakage-free behavioral model predicts that non-return at ROC-AUC 0.82 from week-one signals, with recency, category breadth and active days as the dominant, mutually corroborated drivers; and the modeled value exposure of \$572k is 98% concentrated in two behavioral personas covering ~10% of users, while the highest-churn half of the base carries almost none of it.

A calibrated probability, a customer value proxy, and an explicit EV rule - $p \times \text{uplift} \times \text{value} - \text{cost}$ turn scores into a defensible contact list that concentrates spend where risk and value co-locate, and an equally explicit suppression rule that stops spend where they do not.

Incremental uplift, contact cost, offer economics, out-of-time stability, and every causal claim implicit in the recommendations. None of the projected dollar outcomes are realized results.

Recommended next action is to run the A/B test of the tiered cart-recovery nudge on the EV-selected high-value list, with the fixed analysis date and guardrails as specified, and let its measured lift replace the assumed 12% before any budget scales.

APPENDIX A

Glossary of Terms & Abbreviations

Every term below appears in this report. Definitions are written for a non-technical reader.

A. Model & evaluation

Churn	Here: a customer active in October who records no activity at all in Nov-Dec (e.g. non-return within 60 days. Not a subscription cancellation)
ROC-AUC	Given one churning and one returner, how often the model scores the churning higher. 0.5 = coin flip, 1.0 = perfect. Ours: 0.82.
PR-AUC	Precision-recall area. Its baseline is the prevalence of the positive class (0.771 here), not 0.5, which is why ROC-AUC is the fairer headline.
CV (cross-validation)	Splitting the data 5 ways and rotating train/test, to confirm a result is stable (± 0.001) rather than one lucky split.
HGB	HistGradientBoosting, a gradient-boosted tree model, used as the predictive benchmark against the interpretable logistic regression.
Permutation importance	Shuffling one feature at random and measuring how much the model degrades. A big drop means the feature matters.
Calibration / Brier score	Checking that a stated “80% risk” really means ~80 of 100 such customers churn; Brier measures that (0.129 here, lower is better).
Data leakage	Accidentally giving the model information from the future it is trying to predict. Avoided here by taking features from October and the outcome from Nov-Dec.

B. Experimentation

A/B test	Randomly split customers, treat one group, compare outcomes. This report presents a design and simulation, not a completed experiment.
pp (percentage point)	The absolute gap between two percentages. “+3pp” on a 22.9% return rate means 22.9% $\rightarrow$ 25.9%, not 22.9% $\times$ 1.03, the term exists to avoid that ambiguity.
MDE	Minimum Detectable Effect - the smallest improvement the experiment is powered to detect (+3pp here). A smaller MDE requires many more users.
Power (80%)	The chance of detecting a real effect if one exists. Together with the MDE, it determines the required sample size.
Guardrail metric	A metric watched to ensure a change does no harm elsewhere (here: notification opt-out rate and revenue per user).

Peeking	Checking results repeatedly and stopping once they look significant. It inflates false positives from ~5% to ~14%, so sample size is fixed in advance.
PSM / DiD	Propensity-score matching / difference-in-differences - quasi-experimental methods for estimating causal effects without a randomized test.

C. Business & data

Conversion funnel	The stages a shopper passes through: view → cart → purchase. Fewer people reach each stage, therefore called “funnel”.
Session	One visit to the site. The median user here has exactly one, meaning most shoppers never form a habit.
Persona / segment	A group of customers who behave alike (e.g. “efficient repeat buyers”, “stale / disengaged”).
Revenue at risk	Churn probability × number of users × average spend. The measure that shows where money, not just customers, is being lost.
EV (expected value)	Profit expected from contacting a customer: $P(\text{churn}) \times \text{uplift} \times \text{customer value} - \text{cost}$. The targeting rule contacts someone only when this is positive.
Uplift	Share of would-be churners actually retained because they were contacted, it is the incremental effect of the action, not the churn probability.
CLV	Customer Lifetime Value - total value a customer brings over time. This report uses October spend as a proxy for it.
CRM	Customer Relationship Management system - would hold the churn risk score and trigger campaigns.
RFM	Recency, Frequency, Monetary - the classic three-axis view of customer value used for segmentation.